

Walmart Retail Sales Forecasting

Overview

Retail demand forecasting helps stores optimize inventory, staffing, and promotions. This project predicts Walmart weekly department sales using machine learning models trained on historical sales, economic indicators, and promotional data. The system compares normalized and unnormalized pipelines using Random Forest, Linear Regression, and XGBoost models.

Results

Group 1 (Pipeline A – Raw RF)



- Random Forest on raw features achieved the best performance with the highest accuracy ($R^2 = 0.942$)
- Most effective at capturing non-linear relationships in sales data.

Group 2 (Pipeline A – Raw XGBoost)



- XGBoost on raw features performed strongly with $R^2 = 0.897$.
- Provides a good balance between accuracy and training speed.

Group 3 (Pipeline B – Normalized RF)



- Normalization had almost no effect on Random Forest performance ($\Delta R^2 < 0.001$).
- Tree-based models are scale-invariant by nature.

Group 4 (Pipeline B – Normalized LR)



- Linear Regression underperformed compared to tree-based models with $R^2 = 0.612$.
- Struggles to capture complex non-linear sales patterns.

Methods

Data Preparation

- Missing value handling
- One-hot encoding
- Outlier capping
- Feature engineering

Dual Pipelines

Pipeline 1

- Normalization
 - Random Forest
 - Linear Regression

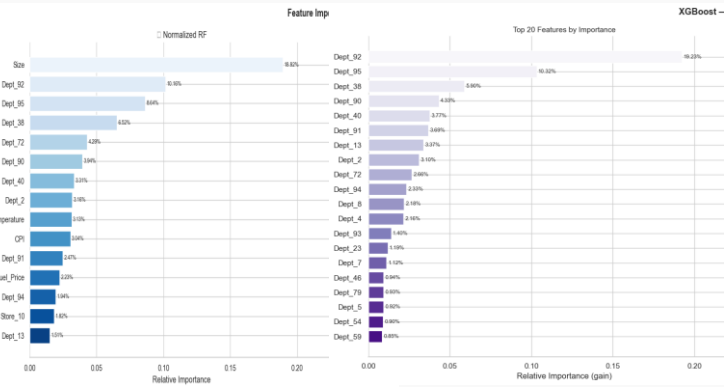
Pipeline 2

- Raw Features
 - Random Forest
 - XGBoost

Evaluation

- MAE
- RMSE
- R^2 Score

Feature Importance



Model	MAE	RMSE	R ²
Random Forest	1842	2963	0.942
XGBoost	2215	3518	0.897
Linear Regression	5621	7834	0.612

Conclusion

This project demonstrates that tree-based ensemble models are highly effective for retail sales forecasting. Random Forest achieved the best predictive accuracy with $R^2 \approx 0.942$, while XGBoost balanced performance and training speed effectively. The study also confirms that normalization has little impact on tree-based methods. Store Size and Department features were the strongest predictors of weekly sales.



Random Forest on raw features achieved the best overall performance with $R^2 = 0.942$, lowest MAE and RMSE.



Normalization had minimal impact on Random Forest, confirming that tree-based models are scale-invariant.



XGBoost performed strongly with $R^2 = 0.897$ and offers faster training with good accuracy.



Linear Regression showed the weakest performance due to inability to capture non-linear sales patterns.



Store Size and Department were the most influential features in predicting weekly sales.